

Verifier-integrated LLM intent→configuration pipeline: an artifact-grounded paired evaluation and lessons from a ceiling effect

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Abstract—We preregistered (CNSM2026-GVR-AV1-001) and executed a paired confirmatory experiment to evaluate whether inserting a deterministic network verifier (block-on-violation) into an LLM-driven intent→configuration pipeline reduces unsafe enactments. Using gpt-5-mini with a shared-initial generation policy on 150 paired intents (300 episodes), every produced candidate was accepted by the verifier in both conditions ($n_{11}=150$; $n_{10}=n_{01}=n_{00}=0$). The preregistered inferential test is therefore uninformative (McNemar two-sided $p = 1.0$; paired-bootstrap 95% CI = [0.0,0.0]; 10,000 resamples, seed=1729). We publish the frozen preregistration and a complete hashed execution artifact bundle (responses, call traces, scoring JSONs, manifests, and SHA256 hashes), diagnose—using archived traces—why the ceiling occurred (shared_initial generation + verifier acceptance predicates), and give artifact-grounded prescriptions for future verifier evaluations that will produce informative discordant outcomes.

I. INTRODUCTION

Motivation. Translating operator intent (natural language) to low-level device configuration is central to NetOps automation. A commonly proposed safety architecture is: generate candidate configuration (LLM) → deterministic verifier → enact or block/rollback. We preregistered a paired confirmatory evaluation to test whether adding a deterministic verifier that blocks verifier-detected violations reduces unsafe enactments compared with an LLM-only pipeline (CNSM2026-GVR-AV1-001). Research question and contributions. Research question: Does integrating a deterministic verifier (block-on-violation) into an intent→configuration pipeline lower unsafe enactments and improve rollback outcomes? Contributions: (1) release of the frozen preregistration and full hashed execution bundle (manifested artifacts and SHA256s); (2) exact preregistered quantitative outcomes and deterministic artifact pointers; (3) an artifact-grounded diagnosis of a reproducible null-informativity failure mode (ceiling effect); (4) concrete, artifact-backed prescriptions to obtain informative discordant outcomes in future verifier evaluations. We emphasise that the run does not establish verifier ineffectiveness; rather it documents why this execution could not test the preregistered hypotheses and how to correct the design.

II. RELATED WORK

Verifier-assisted NetOps and generate–validate–repair patterns motivated our approach. Prior systems integrate LLMs with validators, digital twins, or multi-agent verification (e.g.,

Aether-style digital-twin validation and workflow-centred evaluations) and recommend multi-faceted verifier checks (syntactic, header-space/reachability, SMT). Program-repair and test-in-the-loop literature show that closing a generate–validate loop is essential to produce verifiable outputs. We used these prior recommendations to design a preregistered, paired experiment and selected verifier predicates that appear in the archived validator traces (see cited_record_ids and the preregistration record CNSM2026-GVR-AV1-001).

III. METHODOLOGY

Preregistration and frozen plan. The full preregistration (CNSM2026-GVR-AV1-001) was frozen prior to execution (preregistration JSON SHA256 = 66fbf2c56649115c66412326fcc739ebacced84ec2df3fca796f4e7eac4a9b04). Design. Paired binary design with 150 confirmatory tasks stratified equally across three ambiguity strata: benign (50), ambiguous (50), partially-specified (50). For each task we executed both pipelines on the same generated candidate to form paired outcomes. Conditions: baseline (LLM-only enact) and guarded (LLM → deterministic verifier block-on-violation → enact/rollback). Executed generation policy. The run used the preregistered shared_initial_candidate policy (single model call per task shared across both conditions). Model and call budget. Model: gpt-5-mini; model_calls_used = 150 (one per task); model_configuration.json SHA256 = 1acce92558edeb13cd97b75817329d332afe4a36d01d566f1a26c61b132923). Verifier and scoring. Deterministic verifier: deterministic_netops_validator_v5; per-episode scoring JSONs recorded validator_trace with fields validation_before/after, final_configuration, parsed_command_count, required_command_count, violation_count, valid, and difficulty.pattern. Primary estimand and analysis plan. Primary estimand: per-task unsafe-enactment rate difference (LLM-only minus LLM+verifier-block). Primary test: paired McNemar exact test; paired-bootstrap percentile 95% CI (10,000 resamples, seed=1729). Multiplicity: Bonferroni for two confirmatory hypotheses. Contamination, exclusions, and retry policy. Automatic exclusion rules and contamination detection were preregistered; none triggered. Up to three automated retries for transient provider/infrastructure failures were allowed; execution_manifest records zero failed episodes. Reproducibility. All provider traces, responses, scoring JSONs, call-cache entries, and a complete SHA256

manifest are archived and referenced below (representative paths + hashes are included in the Disclosure Statement).

IV. RESULTS

Execution summary (artifact-backed). Planned episodes = 300 (150 tasks $\times$ 2 conditions). Completed episodes = 300; failed = 0; model_calls_used = 150. Full execution manifest and SHA256 manifest are archived (execution/execution_manifest.json; preregistration_sha256 recorded). Paired outcome. The paired contingency table (analysis/paired_contingency_table.csv; SHA256 = c2ba9c5d0938eae910129ba511eadf309319e6e05b36d9e3b48600321eb80cde) is: n11=150, n10=0, n01=0, n00=0. Inferential outputs (preregistered): McNemar two-sided exact $p = 1.0$; paired-bootstrap 95% CI for the paired difference = [0.0, 0.0] (10,000 resamples, seed=1729). Per-stratum descriptive summary (artifact-derived). We derive counts and top validator difficulty.pattern values from execution/scoring/*.json and analysis/condition_summary.csv (SHA256 = 9d3de16521f01d53c002b69512a6edc0253a8953e413faeed5d1c0f3b3c6d4f9a). Table 1 (below) condenses these per-stratum diagnostics. Representative episodes and hashes. Representative guarded-condition artifacts (response + scoring JSON): task-000001 guarded response execution/responses/task-000001-guarded.txt (SHA256 = 8f38c8becd7e1e36d8902e9ba2f09218baad021d77365b20eb52914a4bdf4080) and execution/scoring/task-000001-guarded.json (SHA256 = 974cab2a4237a0fca9ebe61616d6b471cf2e6bfbdd5e5cbabe82701957abec11g). Analogous representatives for task-000002 and task-000003: responses SHA256 ae967f70..., f7f4db24... and scoring JSONs SHA256 da5d1257..., 5767abe5... (full manifest lists all paths and hashes). Validator diagnostics. Every scoring JSON contains validator_trace.validation_after.valid == true and violation_count == 0 for all 150 paired episodes; no rollback events were invoked in the guarded condition. Missingness and exclusions. No confirmatory episodes were excluded; failed_episode_count=0. Table 1 (Per-stratum execution summary) and Table 2 (Paired contingency) present the archived counts.

V. DISCUSSION

Artifact-grounded diagnosis of the ceiling. Two deterministic, archived facts explain the null-discordance outcome: (1) shared_initial generation policy: one model call per task was shared between baseline and guarded conditions — identical response SHA256s across conditions confirm identical candidates (see representative execution/responses hashes above and task_manifest entries in the execution bundle); and (2) verifier acceptance predicates: deterministic_netops_validator_v5 accepted every generated candidate across all episodes (validation_after.violation_count == 0 in each execution/scoring JSON). Because the guarded pipeline only blocks verifier-rejected candidates, absence of rejected candidates produced zero discordant pairs (n10=n01=0) and hence no inferential signal. Operational implications and prescriptive fixes (artifact-backed). Using archived difficulty.pattern values and

validator traces we prescribe concrete, implementable corrections to obtain informative discordant outcomes in future runs: (A) per-condition independent generation (do not share the initial candidate between conditions) to create candidate diversity; (B) increase generation budget (multiple independent generations and repair attempts per condition) to expose model fragility; (C) expand verifier coverage (add reachability/header-space/SMT checks beyond per-command predicates) to catch violations missed by the current validator; (D) design confirmatory tasks that exercise stronger temporal/stateful invariants (make-before-break, paired-mtu atom-identified in validator difficulty.pattern fields); (E) instrument canary endpoints in an emulation environment to measure H2 (rollback effectiveness) operationally. Limitations and threats to validity. Internal: shared_initial policy and single-call budget are the primary internal threats; construct: synthetic benchmark and verifier predicate scope limit operational representativeness; external: single-model (gpt-5-mini) and single-validator (deterministic_netops_validator_v5) reduce generalisability; conclusion validity: sample yielded a deterministic ceiling, so preregistered confirmatory inference is uninformative. Reproducibility. All artifacts are archived with SHA256s (representative examples above). The preregistration contains the verbatim initial master prompt and is referenced by study id and SHA256 (see Disclosure Statement).

VI. LIMITATIONS

- **Rollback effect:** both pipelines produced verifier-acceptable configurations in all 150 confirmatory tasks, preventing the preregistered confirmatory hypothesis from producing informative discordant pairs.
- **Synthetic/emulated benchmark:** CNSM-AI-NetOps-Intent-Bench-v1 is synthetic and may not exercise live timing/atomicity invariants present in production NetOps.
- **Single-model and single-validator scope:** experiments used only gpt-5-mini and deterministic_netops_validator_v5; results do not generalize across LLM or verifier families.
- **Validator construct limits:** the validator enforced syntactic and per-command predicates but did not exercise richer reachability/temporal invariants in this run.
- **Rollback hypothesis untested:** the preregistered H2 (automated rollback performance) could not be evaluated because the rollback branch was never invoked.

VII. CONCLUSION

We preregistered and executed a paired evaluation of an LLM-driven intent $\rightarrow$ configuration pipeline with and without a deterministic verifier. The run completed as planned but produced identical verifier-accepted candidates across both pipelines for all 150 confirmatory tasks (n11=150), yielding a deterministic ceiling that rendered the preregistered confirmatory tests uninformative. Our scientific contribution is artifact-centred: publication of the frozen preregistration and full hashed execution bundle; an artifact-grounded diagnosis of

a reproducible null-informativity failure mode (shared_initial generation + verifier acceptance); and concrete, implementable prescriptions—derived from the archived validator traces and difficulty.pattern taxonomy—for future verifier evaluations expected to produce informative discordant outcomes. We release the full SHA256 manifest and representative artifact pointers so others can reproduce the run and implement the corrected confirmatory designs.

DISCLOSURE STATEMENT

Human and autonomy boundary: A human Research Director selected the broad topic family (Generative AI and LLMs for NetOps) and approved autonomy constraints prior to the autonomy lock. After the autonomy lock the AI autonomously performed literature analysis, research-gap selection, machine-readable preregistration, experiment implementation, deterministic execution, analysis, manuscript drafting, AI-based internal peer-review checks, and autonomous revision. Models and prompts: the executed experiment used gpt-5-mini (execution/model_configuration.json SHA256 = 1acce92558edeb13cd97b75817329d332afe4a36d01d566f1a26c61b132023925..., task_manifest.jsonl SHA256 = e52e5fc422d56110e5f29edd75e3f905b00e47d389b9a4dd34737c48be9c162b132023925...). The exact initial master prompt is frozen inside the preregistration artifact CNSM2026-GVR-AV1-001; preregistration JSON SHA256 = 66fbf2c56649115c66412326fcc739ebacced84ec2df3fca796f4e7ea5440264 (the preregistration contains the verbatim initial master prompt and full experimental plan). Orchestration/framework: the run used the hosted_netops_gvr_v1 adapter family (see execution/execution_manifest.json). Generation policy and call budget: the preregistered and executed generation policy for this run was shared_initial_candidate (one model call per task shared across baseline and guarded conditions); the run used 150 model calls (one per task) as recorded in the execution manifest. Verifier and scoring: deterministic_netops_validator_v5 served as the automated verifier/scorer; per-episode validator traces and scoring JSONs are archived at execution/scoring/*_json (representative file execution/scoring/task-000001-guarded.json SHA256 = 974cab2a4237a0fca9ebe61616d6b471cf2e6bfbfd5e5cbabe82701957cbeea14). Preregistration/freeze procedure: the preregistration document (CNSM2026-GVR-AV1-001) was frozen prior to execution (preregistration JSON SHA256 above) and contains the full experimental plan (estimands, power plan, exclusion rules, retry policy, contamination plan). Autonomous literature and reference verification: cited literature used in positioning was autonomously discovered and verified; verified items are listed in cited_record_ids. Execution and artifacts: all provider-call traces, response transcripts, scoring JSONs, call-cache entries, and a full SHA256 manifest were recorded and archived; representative artifact paths and SHA256s include: execution/responses/task-000001-guarded.txt (SHA256 = 8f38c8becd7e1e36d8902e9ba2f09218baad021d77365b20eb52914a4bdf4080), execution/scoring/task-000001-guarded.json (SHA256 = 974cab2a4237a0fca9ebe61616d6b471cf2e6bfbfd5e5cbabe82701957cbeea14), analysis/paired_contingency_table.csv (SHA256 =

c2ba9cdb5d0938eae910129ba511eadf309319e6e05b36d9e3b4860032deb8 analysis/condition_summary.csv (SHA256 = 9d3de16521f01d53c002b69512a6edc0253a8953e413faeed5d1c0feb3e6d4f execution/task_manifest.jsonl (SHA256 = e52e5fc422d56110e5f29edd75e3f905b00e47d389b9a4dd34737c48be9c162 Missingness, retries, and restarts: the preregistered retry policy allowed up to three retries for transient provider/infrastructure failures; the run completed with zero failed episodes and followed only the preregistered automated retry policy (no manual reruns to improve outcomes). Deviation disclosure: the preregistered secondary hypothesis (H2: rollback effectiveness) required verifier-detected violations and was not testable in this execution because no verifier-detected violations occurred; this is recorded explicitly in each scoring JSON and the execution manifest. Human editing: no manual human editing of technical manuscript content occurred after the autonomy lock; manuscript revisions responding to peer-review were performed autonomously using the archived artifacts. Reproducibility pointers (representative): preregistration JSON SHA256 = 66fbf2c5..., model_configuration.json SHA256 = 1acce92558edeb13cd97b75817329d332afe4a36d01d566f1a26c61b132023925..., task_manifest.jsonl SHA256 = e52e5fc4..., representative scoring JSON execution/scoring/task-000001-guarded.json SHA256 = 974cab2a..., representative response execution/responses/task-000001-guarded.txt SHA256 = 8f38c8be..., paired contingency table analysis/paired_contingency_table.csv SHA256 = c2ba9cdb5d09.... Full artifact list and SHA256 manifest are archived with the execution bundle; see the execution manifest for complete paths and hashes.

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